Chapter 1

Database Management Systems

1.1 Introduction

A database management system (DBMS) is a software system that allows users to define, create, maintain, and control access to a database. It provides an organized way to store, retrieve, and manage data efficiently.

A DBMS acts as an interface between users and the data stored in the database. It ensures data consistency, security, and concurrency control while providing tools for data manipulation and querying.

1.2 Objectives of a DBMS

The main objectives of a DBMS include:

- Efficient storage and retrieval of data.
- Minimization of data redundancy.
- Data integrity and consistency.